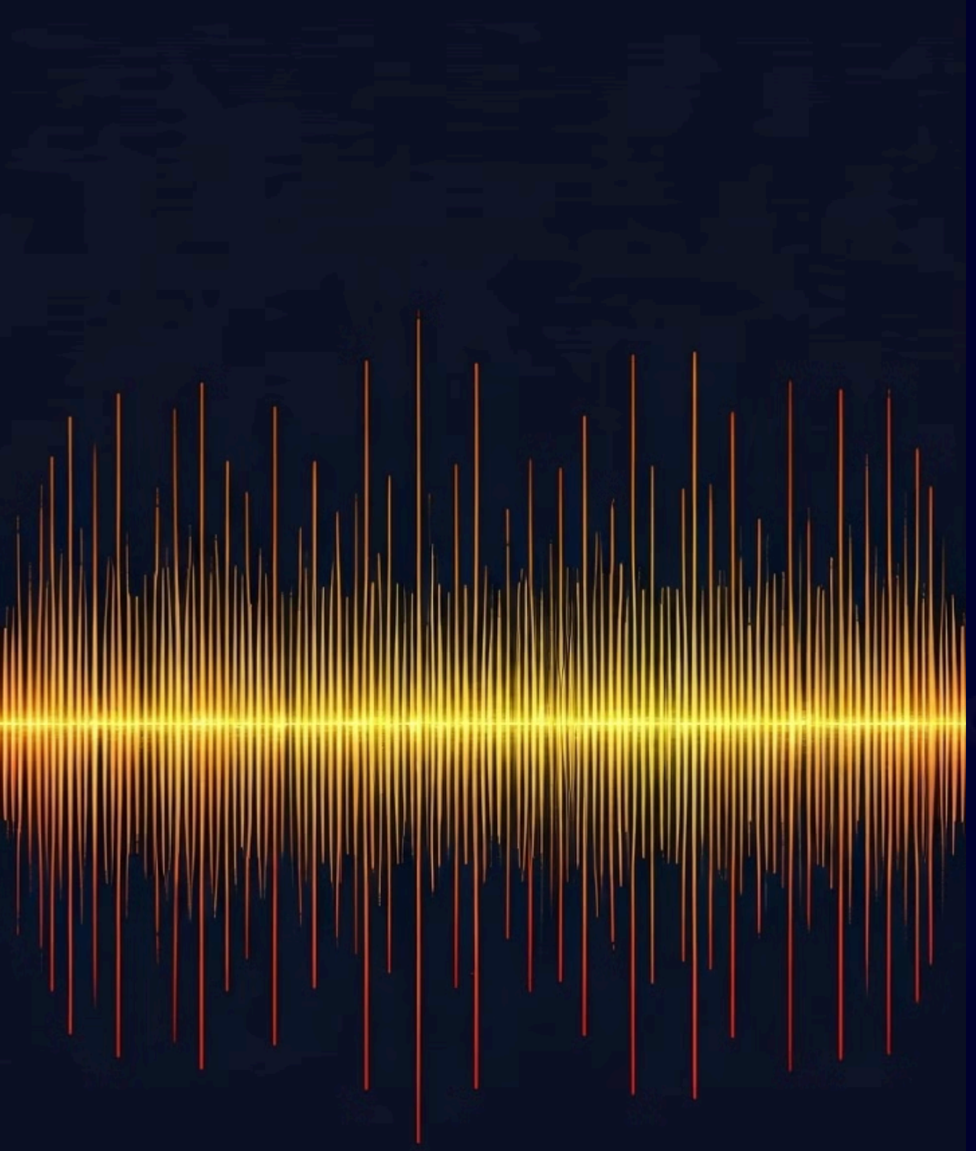


Quadrature Amplitude Modulation (QAM)

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QAM

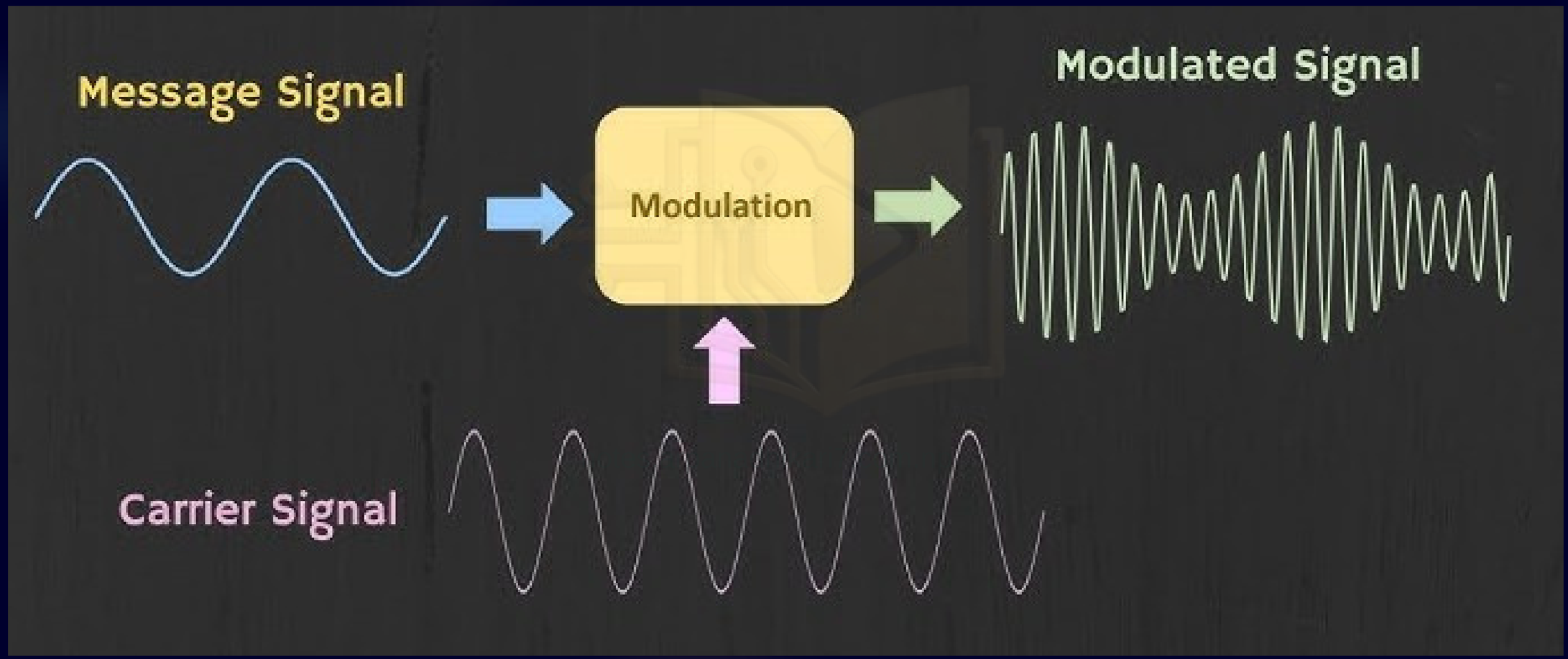
Contents for discussion

- QAM Introduction
- QAM Transmitter
- QAM Receiver
- Advantages of QAM
- Applications of QAM

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Modulation



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Types of Multilevel Modulation

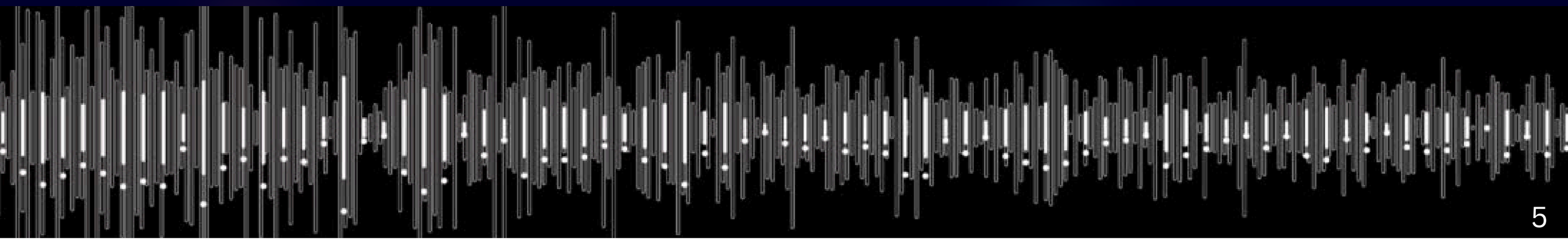


Quadrature Phase Shift Keying (QPSK)

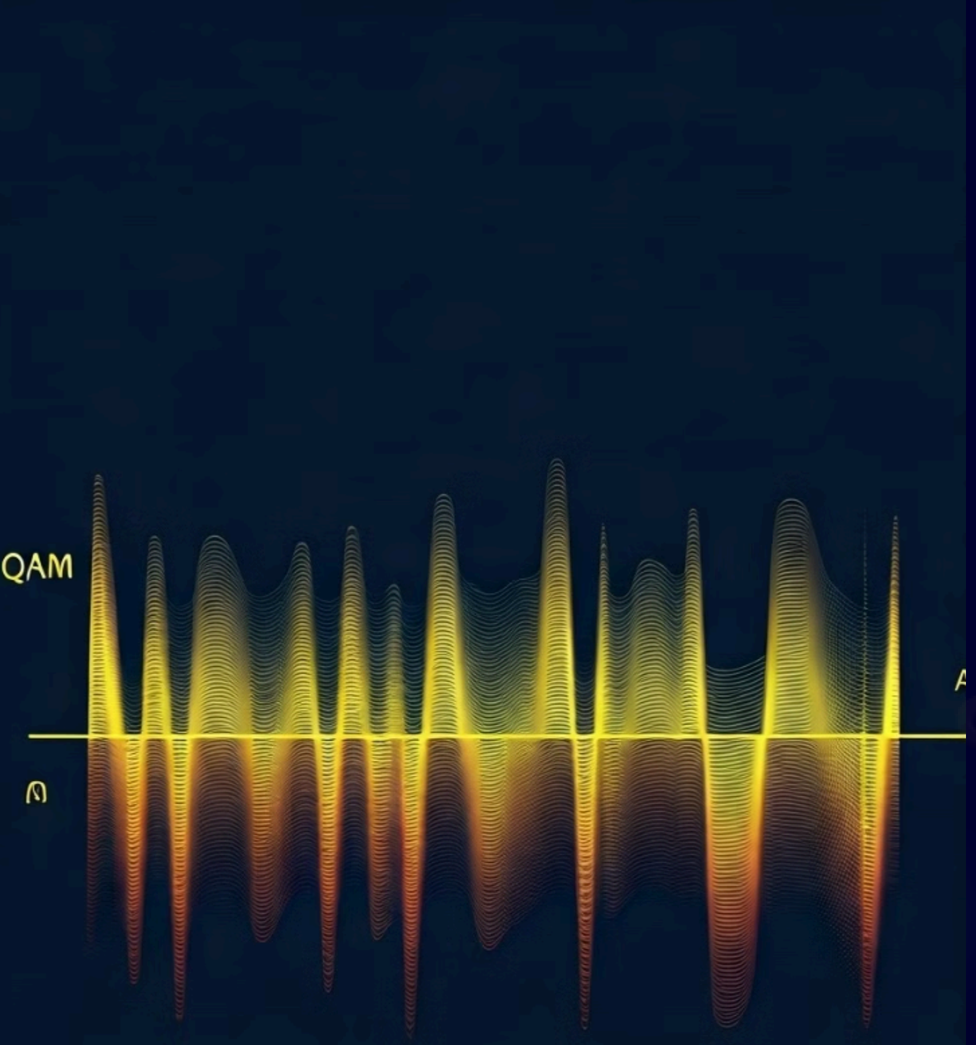
Quadrature Amplitude Modulation (QAM)

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Quadrature Amplitude Modulation (QAM)



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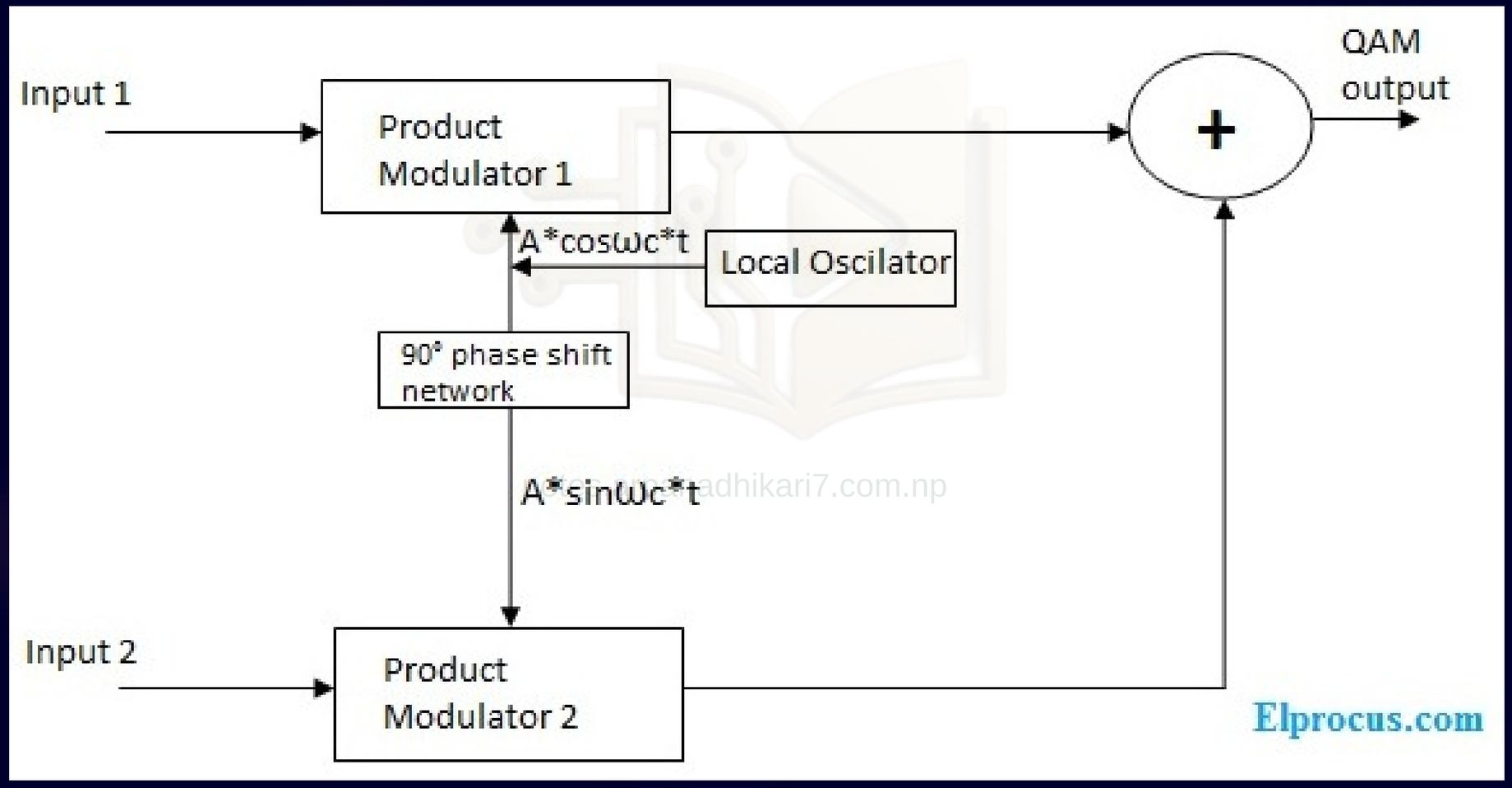
Fundamentals of QAM: Amplitude and Phase Modulation

1. **Amplitude Modulation**
2. **Phase Modulation**

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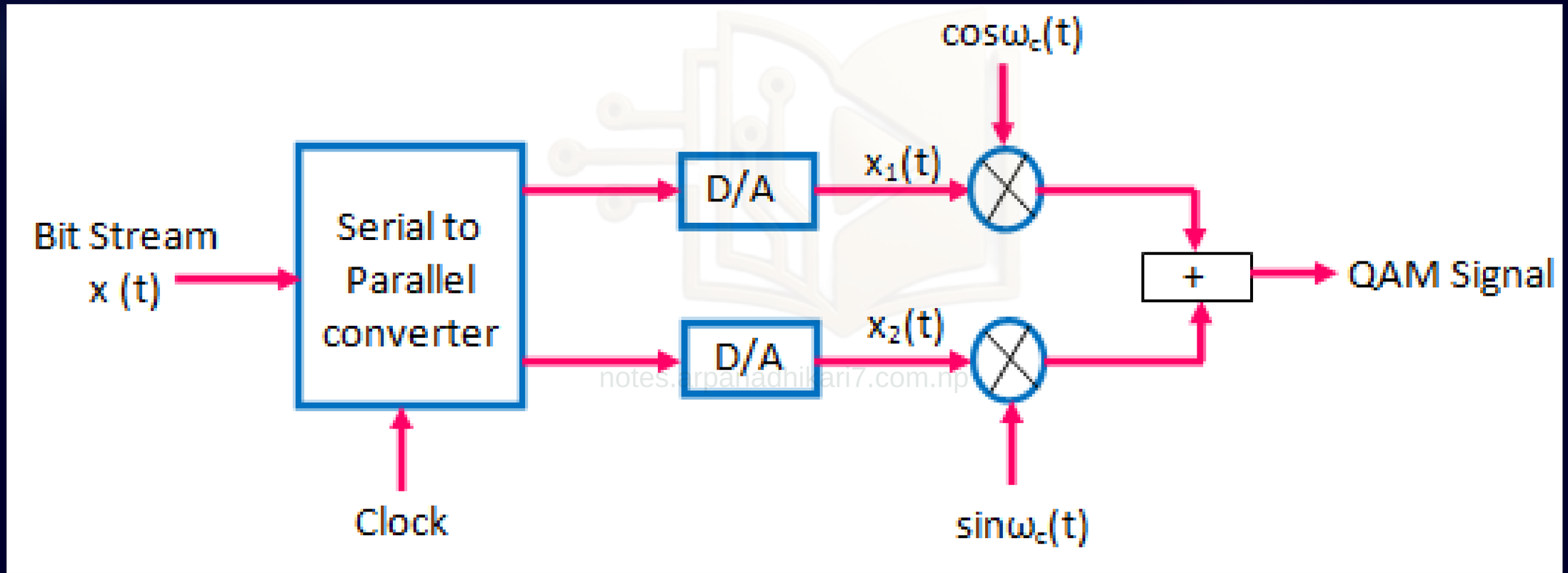
QAM Transmitter



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Digital Signal

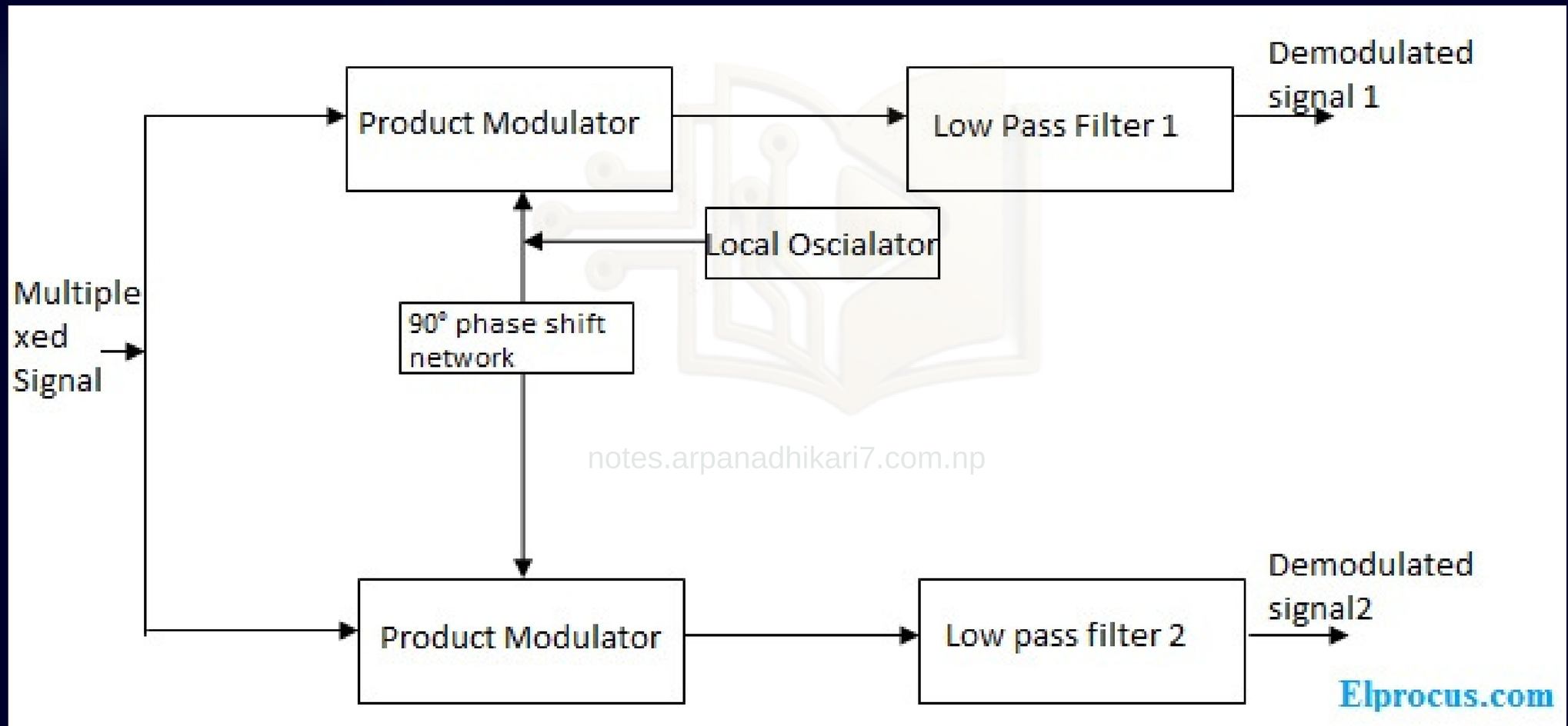
QAM Transmitter



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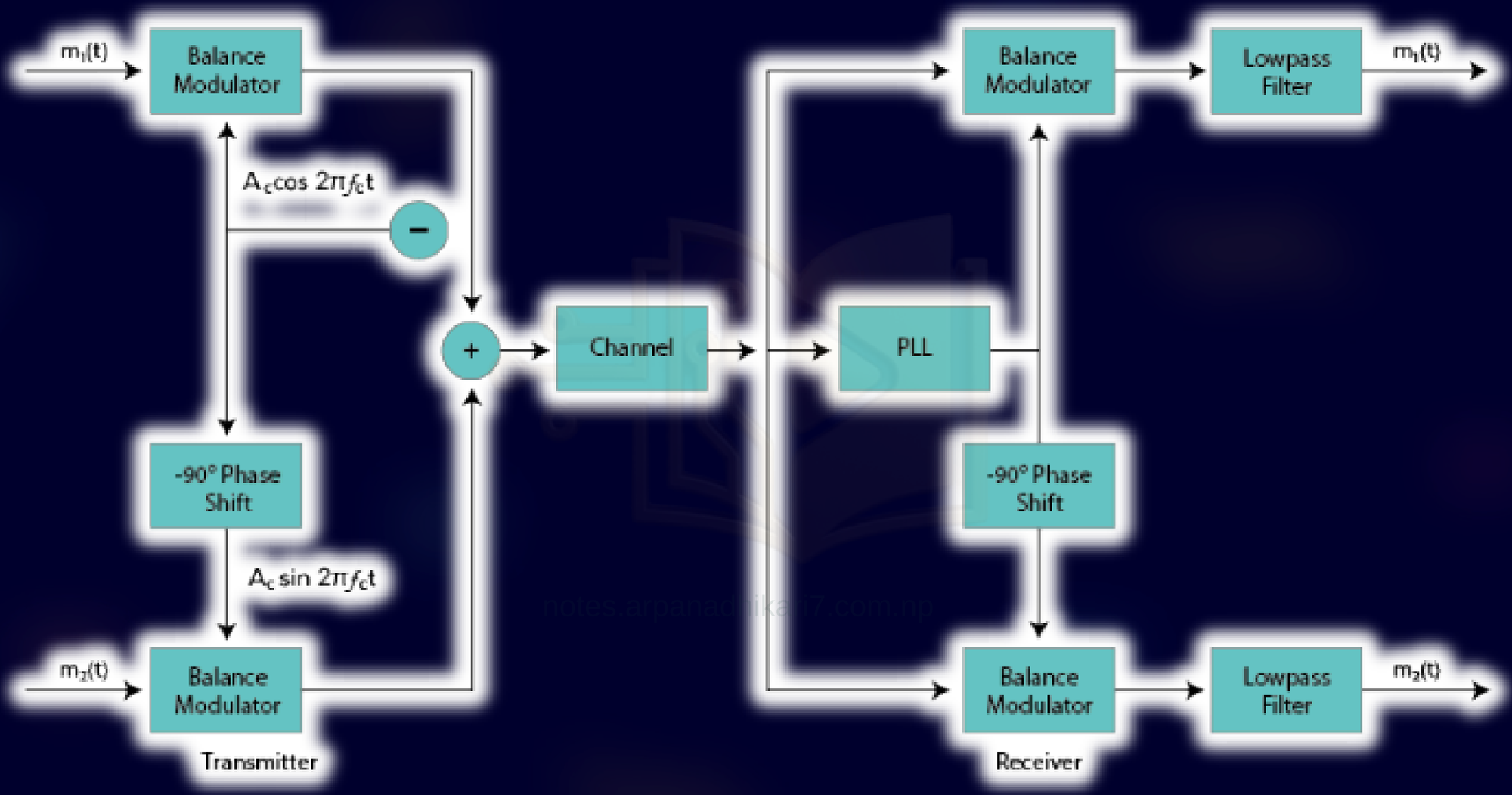
QAM Receiver



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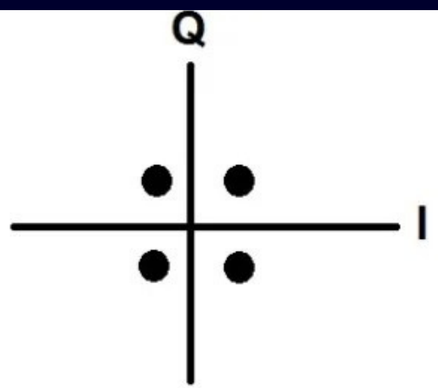
QAM Transmitter and Receiver



PLL = Phase Locked Loop

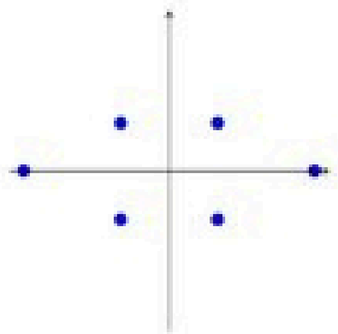
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Types of QAM

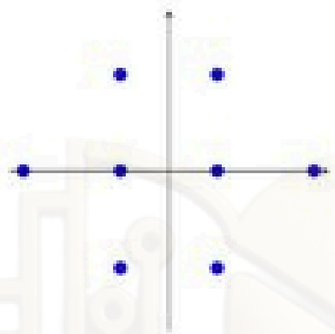


4QAM

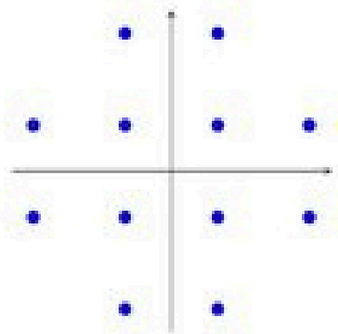
6-QAM



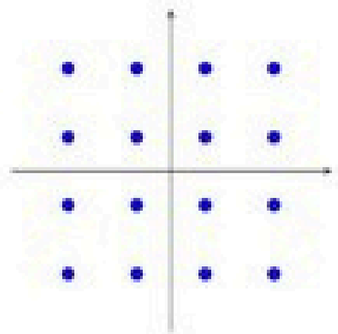
8-QAM



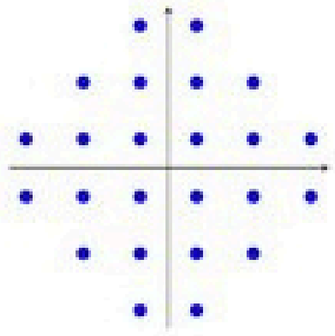
12-QAM



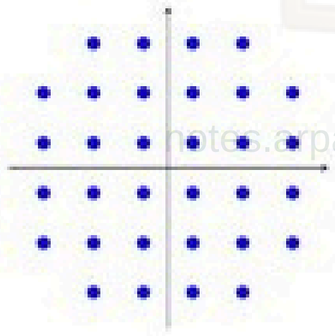
16-QAM



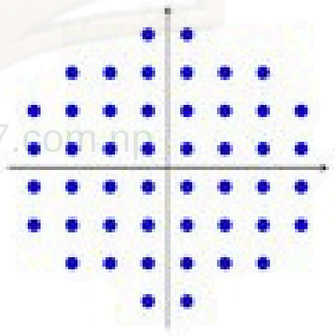
24-QAM



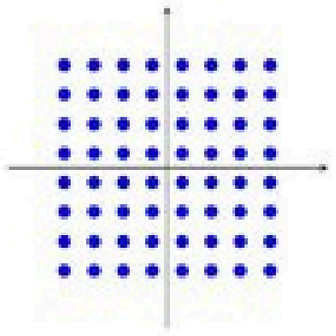
32-QAM



48-QAM

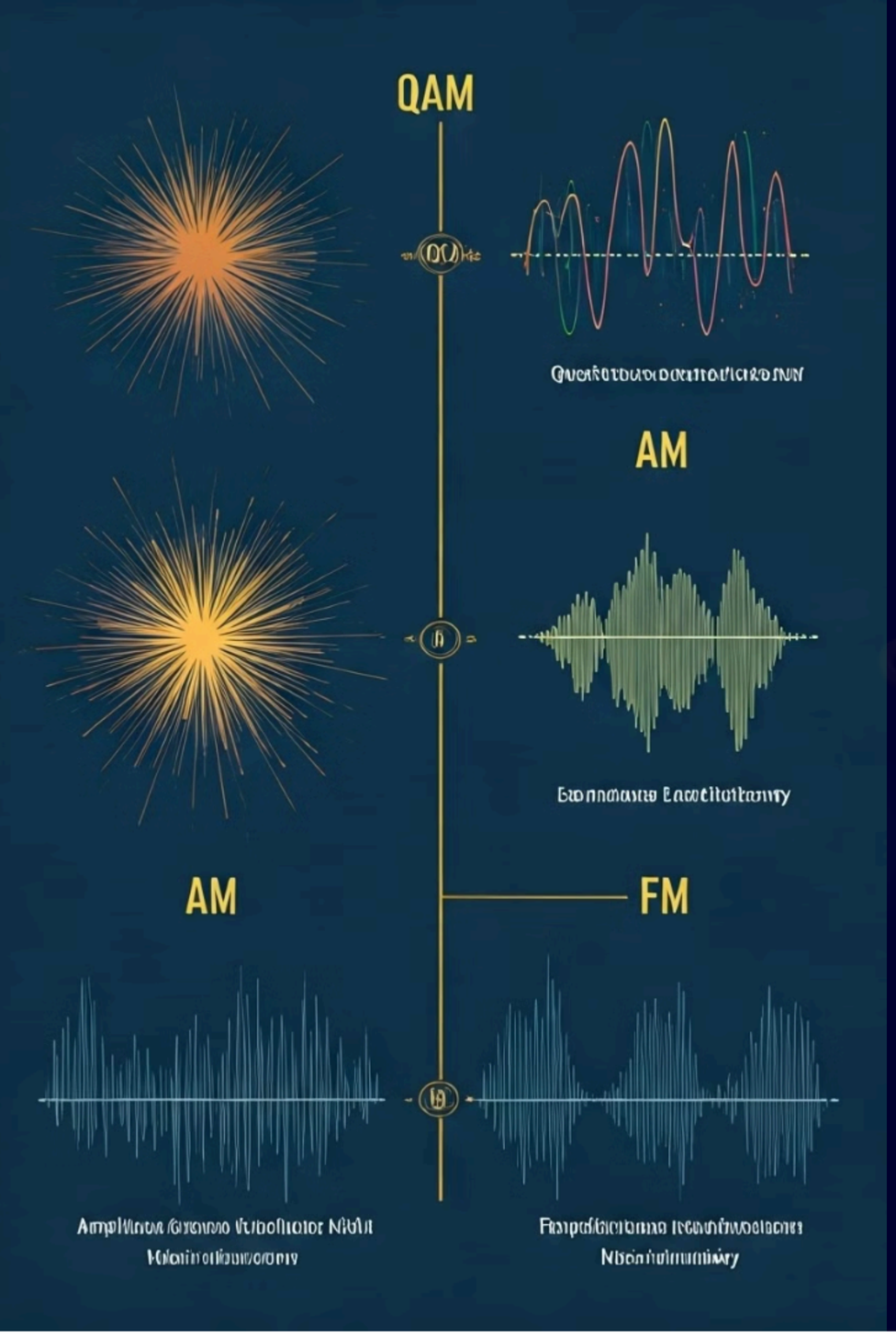


64-QAM



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Advantages of QAM over Other Modulation Techniques

Higher Data Rate

Transmits more data per unit of time

Improved Bandwidth Efficiency

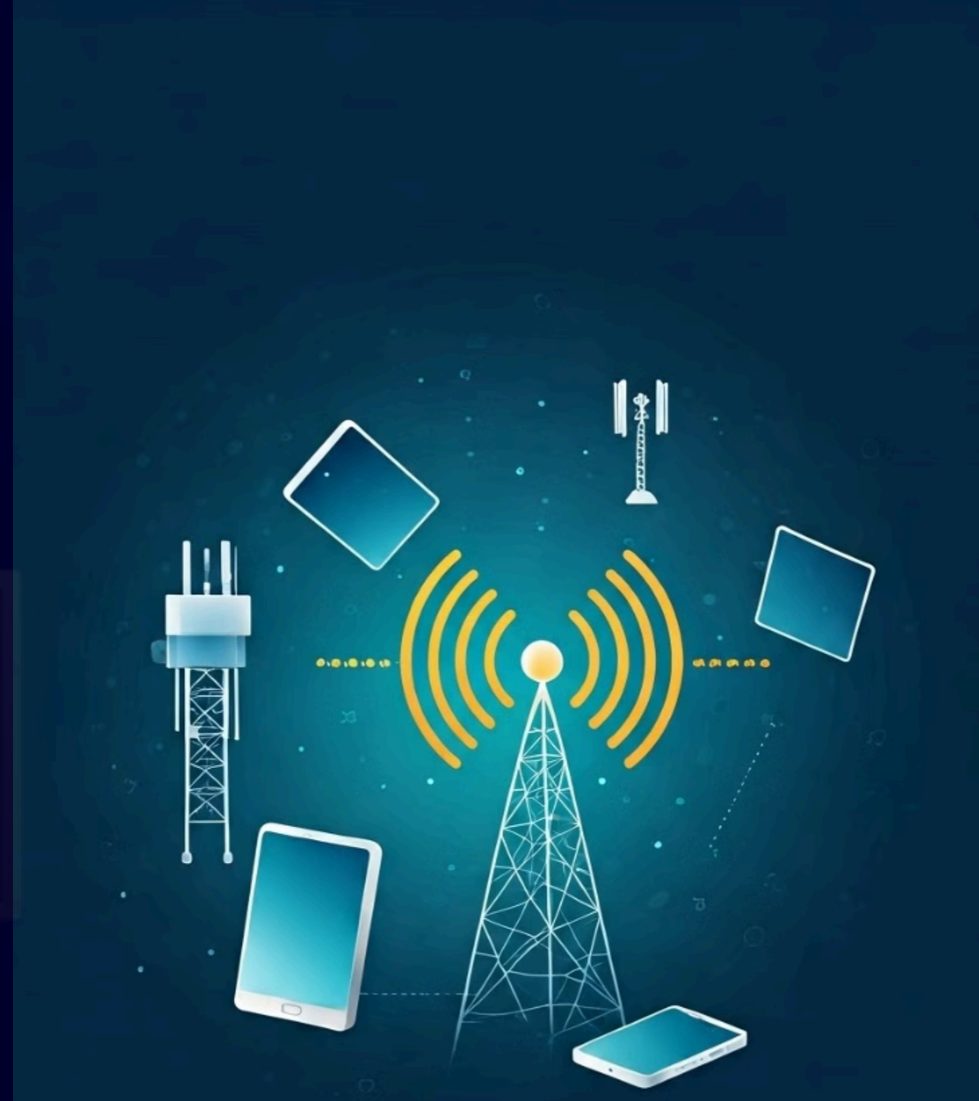
Utilizes available bandwidth more effectively

Robust Performance

Resistant to noise and interference

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QAM in Digital Communications: Cellular Networks and Wi-Fi



Cellular Networks

4G LTE and 5G utilize QAM for high-speed data

Wi-Fi

802.11 standards employ QAM for wireless connectivity

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QAM Applications: Cable TV, Satellite Communications, and Beyond

100M

Cable TV

Digital cable systems use QAM for efficient signal transmission

1G

Satellite Communications

QAM enables high-bandwidth data transmission to remote locations

2G

Digital Broadcasting

QAM facilitates the transmission of high-definition television signals



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